Fiscal Year:	FY 2004	**Task Last Updated:**	FY 11/05/2004
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Project Title:	Space- and Ground-Based Crystal Growth Using a Baffle (CGB)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60616-3729	**Congressional District:**	7
Comments:	Principal Investigator was previously at Rensselaer Polytechnic Institute.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	07/01/1996	**End Date:**	12/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC8-97		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials. Note: Both CGB and SUBSA are funded through NCC8-97.		

Task Description:

At present, the main ground based tasks are: a) Crystal growth experiments with and without the submerged baffle in a multi-zone furnace. The material systems include: doped GaSb, Ge, and InGaSb. b) Numerical simulation of the transport processes in the melt to (i) clarify the results obtained in ground-based experiments and (ii) predict the outcome and optimize the space experiments. The numerical simulations are conducted using computer code NEKTON. c) Design, production and testing of ampoules for SUBSA (ground tests and flight experiments). d) Modification of the SUBSA furnace, to allow ground based crystal growth experiments conducted for the CGB investigation. The modifications include a mechanism for ampoule translation (relative to the furnace and the baffle) and (ii) growth at higher temperatures. Numerical simulation of the experiments on Earth and in Space. It is well established that the perfection of space-grown semiconductor crystals is degraded by (i) motion of free melt boundaries and bubbles and (ii) convection in the melt. These are caused by surface tension and residual acceleration. The main objective of CGB is to develop a method of directional solidification (Bridgman with baffle) which is less sensitive to residual acceleration than the conventional methods (e.g., Bridgman growth, zone melting). This method will be used to reproducibly grow reference quality crystals of doped Ge, GaSb and InGaAs. A key goal is reproducible diffusion-controlled growth in at least 3 subsequent experiments. The shape of the initial transient in composition will be measured and used to determine the diffusion coefficient in the solute layer at the interface, Di. The CGB experiments will demonstrate that Di (which can be measured precisely only in microgravity experiments) is significantly lower than Dbulk measured in shear cells or capillaries. Furthermore, the segregation data obtained in microgravity will be used to test the non-dimensional parameter named the Segregation number.
SUBSA objectives: CGB is the parent investigation to the Materials Science Glovebox (MSG) investigation SUBSA (Solidification Using a Baffle in Sealed Ampoules). The objectives of SUBSA are to: (i) test the “automatically moving baffle” (i.e., baffle driven by melt expansion during freezing); (ii) determine whether the baffle significantly reduces thermal convection in microgravity and (iii) investigate the possible advantages of liquid encapsulation in microgravity conditions. The CGB space experiments will yield reference quality crystals. From the composition of these space-grown crystals, the value of solute diffusivity at the interface Di will be determined. Our ground-based experiments have demonstrated that Di is 3 to 5 times lower than Dbulk measured, in shear cells or capillaries. The exact value of Di can be determined only in microgravity. Solute diffusivity at the interface, Di, is one of the most important transport properties needed for modeling and interpretation of crystal growth experiments. If the experiments yield the expected results, directional solidification with the baffle may become a preferred technique for crystal growth in space. It provides the two key features that are impeding the quality of space-grown crystals: (i) homogeneous starting melt, and (ii) the additional melt stabilization needed to ensure that space experiments crystals are grown under conditions of negligible natural and Marangoni convection. The experiments will demonstrate that the baffle, without additional expense and drawbacks, will reduce the natural (i.e., free buoyancy-driven) convection in the melt to the point that it will not affect segregation.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

CGB is the parent investigation to SUBSA (Solidification Using the Baffle in Sealed Ampoules), a Microgravity Science Glovebox conducted at the International Space Station in 2002. Therefore, this task progress includes some preliminary data obtained in microgravity during the SUBSA investigation. The purpose of this experiments is to duplicate the conditions of the flight experiments conducted in the Microgravity Science Glovebox (MSG) on the International Space Station, except for the level of gravity.
During the past years, 6 out of seven ground control test Sample Ampoule Assemblies (SAA) were prepared following the procedures used to prepare the SUBSA flight ampoules. Using the Mellen furnace at RPI, we have grown several InSb crystals. These crystals were used as seeds and charge for the SUBSA G (SUBSA ground) ampoules. The ground-based crystal growth experiments were performed in the SUBSA ground unit Thermal Chamber at the University of Alabama in Huntsville (UAH). All tests were conducted with the SUBSA thermal chamber in the vertical configuration. In order to duplicate the thermal environment as closely as possible, the maximum temperature of the main heater was adjusted until the position of the solid/liquid interface is identical to that achieved during the flight experiments (approximately 2mm in the seed).

The crystals grown in the SUBSA ground unit are being characterized using:

1. 4 point probe. Because of the extremely low resistance of highly doped InSb crystals, we have developed a new computer controlled 4 point probe setup based on a Keithley 181 digital nano-voltmeter. This setup is now giving reproducible resistance profiles of the grown crystals.

2. Atomic Absorption (AA). Based on the measurements performed so far, flame atomization of Te doped InSb (~ 5x1018 at/cm3) can not produce sufficient resolution and reproducibility. However, the signal produced by flame atomization of Zn doped InSb is sufficiently strong and reproducible. Based on this findings, we are trying to measure the concentration of Te using graphite atomization. Our Perkin-Elmer AA 310 spectrometer with a graphite furnace provides the means to detect metals in quantities as low as 10-12 g.

3. Hall measurements are conducted both at MSFC and RPI. The numerical modeling was focused on Te and Zn segregation at different levels of reduced gravity. We continue using the spectral element code NEKTON. We have managed for the first time to implement a flexible mash, which models the reduction of the melt size during the growth process.

SUBSA Post-Flight Analysis

Post-flight analysis of the SUBSA flight ampoules is being conducted. Computed tomography x-ray scans of the sample ampoules were done at KSC and show the final positions of the baffles within the ampoules. They also demonstrated that SUBSA crystals do not have any internal voids.

The samples were sectioned in a prescribed manner, polished, and prepared for analysis using the Secondary Ion Mass Spectroscopy (SIMS). The SIMS measurements revealed that the 4 Te-doped SUBSA crystals grown in microgravity have nearly identical initial transients. Diffusion controlled end-transient was demonstrated experimentally in one crystal. (SUBSA 02). SUBSA Zn-doped InSb data indicate that systems with k > 1 are particularly suitable for growth in microgravity.

Bibliography Type:	Description: (Last Updated: 05/30/2024)
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Papers from Meeting Proceedings	Spivey RA, Gilley S, Ostrogorsky AG, Grugel R, Luz P, Smith G. "SUBSA and PFMI transparent furnace systems currently in use in the International Space Station Microgravity Science Glovebox." 41st AIAA Aerospace Sciences Meeting & Exhibit, Reno, Nevada, January 2003. 41st AIAA Aerospace Sciences Meeting & Exhibit, Reno, Nevada, January 2003. Paper AIAA 2003-1362. , Jan-2003